Python bytes() Tutorial

bytes() — A built-in sequential data type in Python used for representing immutable sequences of integers in the range **0–255**. It is commonly used to handle binary data such as files, network streams, and encoded text.

■ **Key Characteristics:**
• Homogeneous – accepts only integers (0–255)
• Immutable – cannot be modified after creation
• Supports both positive and negative indexing
• Preserves insertion order
• Iterable using loops

```
# bytes() - Sequential data type
# Sequential data structures support indexing

data = bytes([1, 2, 3, 4, 5])  # Homogeneous (0–255)
print(data)

# Extracting using index
print(data[0])
print(data[1])
print(data[2])
print(data[3])
print(data[4])

# Iterating bytes
for value in data:
    print(value)
```

■ **Use Cases:**
• Handling binary data from files or network streams
• Working with encoded data (UTF-8, ASCII)
• Reading byte-oriented data in IoT and embedded systems

Note: If you need a *mutable* version of bytes, use **bytearray()**.